Exponential and logarithmic equations

Problem 1: Change bsae formula

Use the change base equation to give an exact numeric value using natural logarithm of $\log_4\left[3e^7+2\right]$.
Recalling the base change formula,

$$\log_a[x] = \frac{\log_b[x]}{\log_b[a]}. \tag{1}$$

Now we identify that $a = 4$, $b = e$ and $x = 3e^7 + 2$. Replacing does values into the base change formula we get,

$$\log_4\left[3e^7+2\right] = \frac{\ln\left[3e^7+2\right]}{\ln[4]}$$

$$\boxed{\log_4\left[3e^7+2\right] = \frac{\ln\left[3e^7+2\right]}{\ln[4]}}$$

Problem 2: Solving exponential equations

Solve the following equation

$$3^{2x} - 6\cdot 3^x = 0.$$

We start by re-writting the expression with the help of the following property $a^{mn} = (a^m)^n$,

$$(3^x)^2 - 6\cdot 3^x = 0.$$

Now, let's apply the following change of variable $a = 3^x$. Hence, we can write the equation as,

$$a^2 - 6\cdot a = 0.$$

Now we have a quadratic equation. We can apply the quadratic formula $a = -b \pm \sqrt{b^2-4ac}/(2a)$ to solve the quadratic equation. However, since $c = 0$, we can solve it very simple as follows,

$$\begin{aligned} a^2 - 6\cdot a &= 0 \\ a^2 &= 6\cdot a \\ \frac{1}{a}\cdot a^2 &= \frac{1}{a}\cdot 6\cdot a \\ a &= 6. \end{aligned}$$

Now that we have a numeric value for a, we can undo the change of variable and solve for x,

$$\begin{aligned} a &= 6 \\ 3^x &= 6 \\ \log\left[3^x\right] &= \log\left[6\right] \\ x\log\left[3\right] &= \log\left[6\right] \\ x &= \frac{\log\left[6\right]}{\log\left[3\right]} \end{aligned}$$

$$\boxed{x = \frac{\log\left[6\right]}{\log\left[3\right]}}$$

Note: See how the answer is actually the change base formula applied to $\log_3\left[6\right] = \frac{\log[6]}{\log[3]}$. This makes sense, since $\log_3\left[3^x\right] = x$. Therefore, if you choose to applied $\log_3$ instead of log you will get the same answer. However, to give a numeric value, you need to be able to use the base 3 in your calculator. If you dont have that option, then use the change base formula answer.

Problem 3: Solving logarithmic equation

Solve the following equation by finding the values of x which the equation os true.

$$\log_6\left[2x+2\right] + \log_2\left[2x-2\right] = 4.$$

Let's start by reducing the logarithmic terms with the properties of logarithms and then solv for x.

$$\begin{aligned} \log_6\left[2x+2\right] + \log_2\left[2x-2\right] &= 4 \\ \log_6\left[(2x+2)(2x-2)\right] &= 4 \\ 6^{\log_6\left[(2x+2)(2x-2)\right]} &= 6^4 \\ (2x+2)(2x-2) &= 6^4 \\ 4x^2 - 4x + 4x - 4 &= 6^4 \\ 4x^2 - 4 &= 6^4 \\ 4x^2 &= 6^4 + 4 \\ x^2 &= \frac{6^4+4}{4} \\ x &= \pm 5\sqrt{13}. \end{aligned}$$

Now let's replace the numeric value into the original equation to see if both solutions are valid. We start with the positive one,

$$\begin{aligned} \log_6\left[2\cdot 5\sqrt{13}+2\right] + \log_2\left[2\cdot 5\sqrt{13}-2\right] &\approx 2.030990 + 1.969009 \\ &\approx 3.999999 \end{aligned}$$

So this solution is valid. Now let's try the negative solution,

$$\begin{aligned} \log_6\left[2\cdot -5\sqrt{13}+2\right] + \log_2\left[2\cdot -5\sqrt{13}-2\right] &= \log_6\left[-34.05\right] + \log_6\left[-38.05\right] \\ &= \text{Math error} + \text{Math error}. \end{aligned}$$

Since the logarithmic function is not defined for negativa numbers $x < 0$, the second value is not a solution of the equation.

$$\boxed{x = 5\sqrt{13}}$$

Problem 4: Earthquakes

An earthquake of magnitude 8.5 was detected in the coastline of Michoacan. Recalling that the energy releases by an earthquake is $\log[E] = 4.8 + 1.5M$. Express the answers in tones of TNT (1tone of TNT = 4.184×10^9).

- What is the release energy in tones of TNT?
- What is the difference energy with an earthquake of magnitude 8.2?

First we solve for the energy,

$$\log[E] = 4.8 + 1.5M \to E = 10^{4.8+1.5M} = 10^{4.8} \cdot 10^{1.5M}.$$

Now we can find the release energy in tones of TNT for an earthquake of magnitude 8.2.

$$\begin{aligned} E_{8.5} &= 10^{4.8} \cdot 10^{1.5 \cdot 8.5} \\ &= 10^{4.8} \cdot 10^{12.75} \\ &\approx 3.548\,133 \times 10^{17}. \end{aligned}$$

Now let's express this number in tones of TNT by dividing the result over 4.184×10^9.

$$\begin{aligned} E_{8.5} &\approx \frac{\approx 3.548\,133 \times 10^{17}}{4.184 \times 10^9} \\ &\approx 84\,802\,413.96. \end{aligned}$$

$$\boxed{E_{8.5} \approx 84\,802\,413.96}$$

Now let's compute the difference energy between the earthquake of 8.5 with 8.2,

$$\begin{aligned} E_{8.5} - E_{8.2} &= 10^{4.8} \cdot 10^{1.5 \cdot 8.5} - 10^{4.8} \cdot 10^{1.5 \cdot 8.2} \\ &\approx 3.548\,133 \times 10^{17} - 1.258\,925 \times 10^{17} \\ &\approx 2.289\,208 \times 10^{17}. \end{aligned}$$

Now let's express the difference in terms of tones of TNT,

$$\begin{aligned} E_{8.5} - E_{8.2} &\approx \frac{2.289\,208 \times 10^{17}}{4.184 \times 10^9} \\ &\approx 54\,713\,384.32 \end{aligned}$$

$$\boxed{E_{8.5} - E_{8.2} \approx 54\,713\,384.32}$$

Exercise 1 Suppose that you invest \$500 at a annual interest rate of 1.5%, compounded monthly.

- How long will it take for your money to be \$10 000?
- How long will it take for your money to be \$10 000 if the rate is 3%?

Exercise 2 The sound level β in decibels (db), with and intensity I is given by $\beta(I) = 10\log[I/I_o]$ Where $I_o = 1 \times 10^{-12}$.

- Calculate the sound level for a pneumatic drill that operates to and intensity of $I_o = 1 \times 10^{-12}$
- Calculate the intensity of the pain threshold of 120 db

Exercise 3 Use the change base equation to give and exact numeric fraction using natural logarithm of $\log_2[2e^2]$

Exercise 4 Solve the following equation $6^{x-3} = 2^{x+2}$.